

Fundamentals of Derivatives Trading Strategies

Group 1: Rolling Futures Strategy

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Contents

1	EXECUTIVE SUMMARY	3
2	STRATEGY DESIGN AND MARKET EXPOSURE	4
3	THEORETICAL FRAMEWORK: PRICING S&P500 FUTURES	6
4	BASLINE STRATEGY: SYSTEMATIC ROLLING	11
5	OPTIMIZED STRATEGY: DYNAMIC ROLLING	14
6	BACK-TEST RESULTS AND PERFORMANCE EVALUATION	17

1 EXECUTIVE SUMMARY

Project Objective

This study focuses on optimizing the rolling window for futures. The first step was to generate a portfolio of the ten largest companies in the S&P 500, enabling the use of S&P 500 futures contracts as hedging instruments. The analysis examines two distinct approaches:

- **Baseline Strategy:** This basic approach is a mechanical process based solely on predetermined rolling dates. Its objective is to isolate the effects of the rolling process, transaction costs, and interest rates.
- **Optimized Strategy:** This approach is more flexible and executes trades only when a statistical indicator, the Basis Z-Score, signals that futures prices deviate from their theoretical fair value.

Summary of Key Findings

Hedging substantially reduced losses during periods of market stress, particularly during the equity selloff in 2022. While higher hedge ratios decreased total returns and reduced the final net asset value, they improved risk-adjusted performance. The highest Sharpe ratio (1.318) and Calmar ratio (0.807) were achieved with greater hedge exposure, attributable to lower volatility and smaller drawdowns.

- **Persistent Market Inefficiencies:** In all tested configurations, the market roll spread consistently exceeded the theoretical fair-value roll spread. This finding indicates that inefficiencies persist in the futures market and are not addressed by a purely mechanical rolling approach.
- **Identification of Market:** The back-test period, spanning late 2021 to early 2026, revealed two distinct market environments. Initially, the market experienced backwardation, resulting in relatively expensive rolls for short positions. From mid-2022 onward, contango became the prevailing regime, generally providing a more favorable roll yield for short hedges.
- **Importance of Risk Management:** Hedging substantially reduced losses during periods of market stress, particularly during the equity selloff in 2022. While higher hedge ratios decreased total returns and reduced the final net asset value, they improved risk-adjusted performance. The highest Sharpe ratio (1.318) and Calmar ratio (0.807) were achieved with greater hedge exposure, attributable to lower volatility and smaller drawdowns.

2 STRATEGY DESIGN AND MARKET EXPOSURE

Although this section serves as the foundation of the research, it is presented concisely because of its simplicity. The primary focus of the study is to detail the methods used to achieve the optimization of the rolling window. To evaluate the rolling strategy, a portfolio of US equities was constructed using the ten largest companies in the S&P 500 by market capitalization, including Apple, Microsoft, Alphabet, Amazon, Nvidia, Meta, Tesla, JPMorgan, Visa, and Johnson & Johnson. This choice allowed us to use S&P 500 futures as a hedging instrument. Data were retrieved and manipulated using Python and the `yfinance` API. The analysis covers a five-year period from late 2021 through early 2026, containing both bull and bear markets.

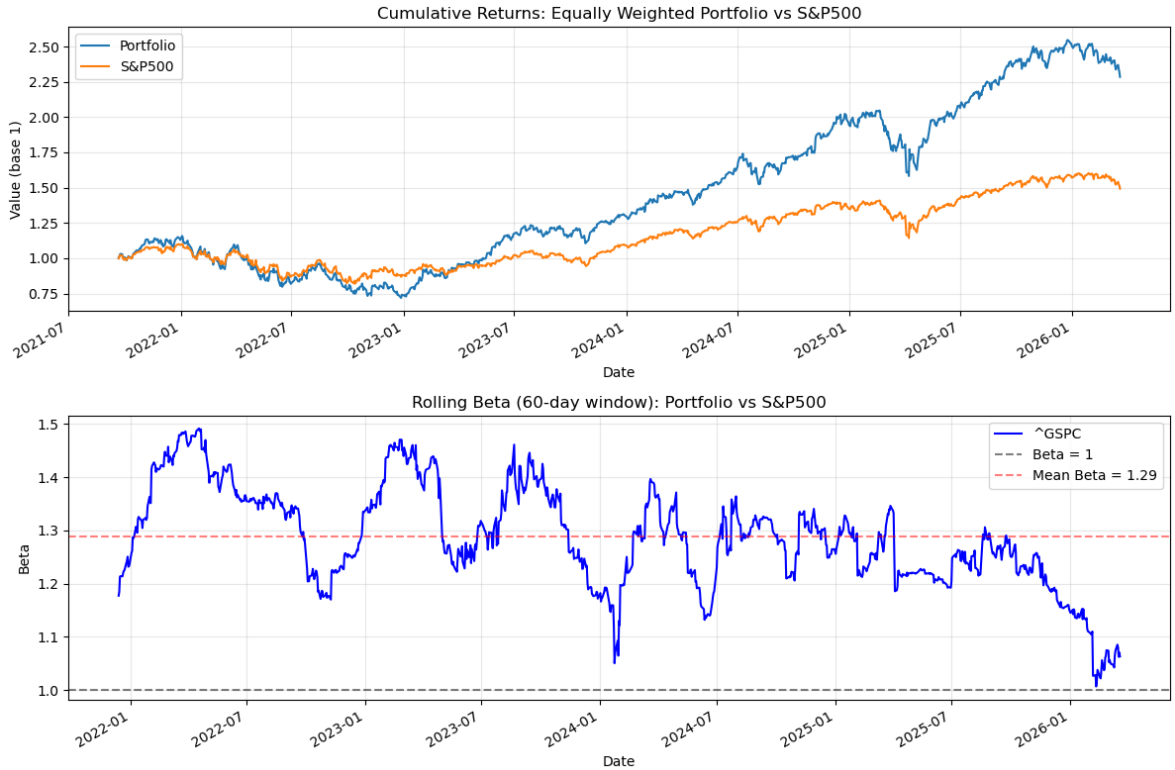
To keep things simple, an equally weighted (EW) approach was implemented, assigning each stock a 10% weight. We used the daily simple return of each stock to compute the total portfolio returns, which is the sum of the product between each stock's weight and its daily returns:

$$R_p = \sum_{i=1}^n w_i R_i$$

Stable market exposure is an essential element of this strategy. A 60-day rolling beta was calculated, as this window is responsive enough to capture sudden market transitions while remaining stable enough to avoid frequent hedge adjustments. This was calculated by examining the rolling covariance between the portfolio (R_p) and the S&P 500 (R_m), divided by the market variance:

$$\beta_{\text{rolling}} = \frac{\text{Cov}(R_p, R_m)}{\text{Var}(R_m)}$$

The results show that the rolling beta fluctuated throughout the test period, reaching a maximum of 1.491 and a minimum of 1.007, with an average value of approximately 1.288. Because the portfolio exhibits a beta greater than 1 (approximately 1.29), it is more sensitive to market movements than the wider index. To reduce this sensitivity to different levels of β , it is necessary to sell S&P 500 futures contracts as a hedge.



Metric	EqualWeight	S&P500 (^GSPC)
Final Portfolio Value	\$225,762,623.95	\$148,021,215.98
Annualized Return	0.21	0.10
Annualized Volatility	0.24	0.17
Sharpe Ratio	0.87	0.60
Max Return	0.12	0.10
Min Return	-0.06	-0.06
Max Drawdown	-0.38	-0.38
Win Rate	0.54	0.53
Skewness	0.26	0.17
Kurtosis	4.76	7.03

Table 1: Portfolio Performance Summary vs. S&P 500 (^GSPC)

While the Equal Weight portfolio delivered a massive \$77 million in alpha over the benchmark, its superior Sharpe Ratio (0.87 vs. 0.60) indicates that this was not just a result of blind luck or taking on more leverage; it was a more efficient use of risk units. However, the identical Max Draw down of -0.38 suggests that during systemic "black swan" events, the Equal Weight strategy offers no structural protection compared to the index. It rides the same roller coaster down, but its higher Skewness (0.26) implies it has a "right-skewed" tendency to capture explosive upside moves more effectively than the market-cap-weighted benchmark. From a behavioral standpoint, the Kurtosis of (7.03) for the benchmark suggests a "fat-tailed" distribution, meaning it is more prone to extreme shocks than the Equal Weight portfolio (4.76).

3 THEORETICAL FRAMEWORK: PRICING S&P500 FUTURES

This section develops the theoretical and empirical foundation upon which the rolling strategy is constructed. Before any roll decision can be optimized or any trading signal designed, a prior question must be answered: what should a futures contract be worth? We address this using the cost-of-carry model, and then describe in detail how each of its inputs was estimated from market data. The resulting outputs: fair values, time-to-expiry curves, interpolated risk-free rates, and basis time-series, serve as the direct inputs to the strategy and back-test described in subsequent sections.

The E-Mini S&P 500 Futures Contract

Our strategy is built exclusively on E-Mini S&P 500 futures (CME ticker root: ES). The E-Mini is the most liquid equity index futures contract globally and is particularly suited to systematic hedging strategies on account of its contract design, deep liquidity, and low transaction costs.

Each contract carries a multiplier of \$50 per index point, implying a notional value of approximately \$250,000-\$300,000 at current index levels. The minimum price fluctuation is 0.25 index points, equivalent to \$12.50 per contract. Contracts are listed on a quarterly cycle, with expiry months in March (H), June (M), September (U), and December (Z), and are traded nearly 24 hours a day on the CME Globex platform. Settlement is cash-based rather than physical: upon expiry, no underlying shares change hands; instead, positions are marked against the official settlement price. Trading terminates at 9:30 a.m. Eastern Time on the third Friday of the contract month. Importantly for cost analysis, no stamp duty or transaction tax is levied on futures trading in the United States, contributing to the overall efficiency of the instrument.

Because each contract has a finite life, a participant wishing to maintain continuous exposure beyond a single contract's expiry must systematically close the expiring position and re-establish it in the next deferred contract. This operation, the roll, is the central focus of the strategy, and its timing and cost depend critically on the relationship between the market price of each contract and its theoretical fair value.

Contract Life-cycle and Data

The back-test covers the period 21 September 2021 to 20 March 2026, encompassing 18 quarterly contracts from ESZ1 (December 2021) through ESH6 (March 2026). Historical closing prices for all expired contracts were sourced from Bloomberg. Daily closing prices for the S&P 500 spot index were obtained from Yahoo Finance. The four SOFR term rates used for interest rate estimation (described in Section "Risk-Free Rate Estimation") were also extracted from Bloomberg. Dividend data is discussed separately in Section "Dividend Yield Estimation".

Each futures contract is active only during a specific window: it begins trading well before its expiry and ceases on the final trading day. This produces a ragged structure in the data: the price matrix has one column per contract, populated only during that contract's active period and left undefined outside it. Contract expiry dates are inferred directly from the data as the last date on which a valid price is observed for each contract, avoiding any hard-coded assumptions about the expiry calendar.

The Cost-of-Carry Model

No-Arbitrage Derivation

The fair value of an equity index futures contract is determined by a no-arbitrage argument. Consider two strategies that both deliver one unit of the index at maturity T :

- **Strategy A (Cash-and-Carry):** Borrow S_0 at the continuously compounded risk-free rate r , purchase one unit of the spot index today, and hold it until maturity T , reinvesting dividends received at yield q back into the index.

- **Strategy B (Synthetic via Futures):** Enter a long futures position at the agreed delivery price F_0 , which requires no upfront cash outlay.

Both strategies deliver one unit of the index at T . If their costs differ, a riskless profit is achievable. If the futures price exceeds fair value, an arbitrageur borrows at r , buys the spot, and shorts the futures, earning the difference at no risk. If the futures price is below fair value, the arbitrageur short-sells the spot, lends the proceeds at r , and goes long the futures. Either deviation is eliminated by competition, forcing the equality:

$$F_0 = S_0 \cdot e^{(r-q) \cdot T}$$

where S_0 is the current spot index level, r is the continuously compounded risk-free rate over horizon T , q is the continuously compounded dividend yield, and T is the time to expiry expressed in years.

Economic Interpretation: Contango and Backwardation

The exponent $(r - q)$ represents the net carry of holding the underlying index rather than a futures contract. Holding the physical index incurs a financing cost at rate r (since capital must be committed) but generates dividend income at rate q . When $r > q$, net carry is positive and futures trade above spot, this condition known as contango. When $q > r$, net carry is negative and futures trade below spot, a condition known as backwardation.

This distinction has a direct and material effect on rolling costs. In a contango market, rolling a short futures position forward (buying back the expiring near contract at a relatively lower price and re-shortening the deferred contract at a higher price) generates a positive spread for the short holder at each roll. In a backwardation market, the reverse is true: rolling becomes costly for a short holder, since the near contract is priced above the deferred contract.

Over the back-test period, two distinct regimes are identifiable. During late 2021 and the first half of 2022, SOFR rates remained near zero while the S&P 500 dividend yield exceeded the short-term rate, placing the market in backwardation. From approximately July 2022 onward, the Federal Reserve’s aggressive rate-hiking cycle pushed SOFR well above the dividend yield, transitioning the market firmly into contango, the dominant regime for the remainder of the sample. Additionally, a yield curve inversion (one-month SOFR exceeding the twelve-month rate) is observed from March 2023, reflecting elevated short-term rate expectations that were not yet fully transmitted to longer tenors.

Time-to-Expiry

The Time-to-expiry (T) is calculated daily for each active contract as the number of calendar days remaining until the contract’s last valid trading date, divided by 365. Formally, for contract c observed on date d :

$$T_{c,d} = \frac{Expiry_c - d}{365}$$

T is defined only during the window in which a given contract is actively traded, and is set to undefined outside that window. The resulting time-to-expiry series for each contract takes the form of a declining ramp that resets to approximately 0.25 years at the start of each new quarterly contract’s active life, producing the characteristic declining ramp pattern associated with a rolling futures program.

Risk-Free Rate Estimation

Data: Term SOFR Rates

Following the global transition away from LIBOR, the Secured Overnight Financing Rate (SOFR) is the standard risk-free reference rate for US dollar instruments. We use CME Term SOFR rates, which represent forward-looking, transaction-based expectations of compounded overnight SOFR over standard tenors. These are conceptually more appropriate than overnight SOFR for discounting over

a multi-week or multi-month futures horizon, as they embed the market’s expectation of the short-rate path over the relevant period.

Four tenor points are observed daily from Bloomberg: 1-month, 3-month, 6-month, and 12-month Term SOFR. All rates are expressed as continuously compounded annual rates. This choice of benchmark is theoretically sound: because E-Mini futures are margined and marked to market daily with no upfront cash exchange, the relevant financing rate is a short-term, near risk-free rate (precisely what Term SOFR measures).

Cubic Spline Interpolation

A central practical challenge in applying the cost-of-carry formula is that the time-to-expiry T for any given contract on any given day almost never coincides with a standard quoted tenor, requiring a principled interpolation of the rate curve. Simple alternatives are inadequate: linear regression does not pass through the observed rates, which are actual borrowing costs that must be matched exactly; piecewise linear interpolation passes through every node but creates kinks, producing inconsistent rate estimates on either side of a tenor boundary; and a single high-degree polynomial through all points fits exactly but oscillates erratically between nodes, generating economically implausible intermediate values.

We therefore employ a *natural cubic spline*, which fits a distinct third-order polynomial to each segment between adjacent tenor nodes, subject to constraints enforcing exact interpolation at every node, continuity of the first and second derivatives at every interior node, and zero second derivative at the two endpoints. The result is a curve that passes through all observed rates, is smooth everywhere, and does not impose arbitrary curvature beyond the observed range. The cubic degree is the minimum that supports two levels of differentiability, the highest level of smoothness achievable while keeping the boundary conditions well-defined and economically justifiable.

In practice, with four SOFR tenor points ($n = 4$), the spline consists of three cubic segments and is fully determined by 12 coefficients, resolved by the constraint system described above. For each contract and each date, the interpolated risk-free rate r is obtained by evaluating the spline at the corresponding T , using the tenor grid $\{1/12, 3/12, 6/12, 1 \text{ year}\}$.

Dividend Yield Estimation

The Role of Dividends in Futures Pricing

Of the three inputs to the cost-of-carry formula, the dividend yield is the only one that cannot be directly observed from market prices without a modeling assumption. The dividend yield relevant to a futures contract with expiry T requires an estimate of aggregate future dividend payments from all 500 S&P 500 constituents over the remaining life of the contract. The quality of this estimate is a direct determinant of the precision of the fair value calculation and, by extension, of the accuracy of the roll signal. A trader with a superior dividend forecast occupies a structural informational advantage when assessing whether a roll spread is rich or cheap relative to fair value.

Data Source: Bloomberg IN060

Dividend estimates are sourced from Bloomberg’s IN060 field (`IDX_EST_DVD_CURR_YR`), which provides the index-level estimated dividends for the S&P 500 for the current fiscal year, computed from the best available estimate for each constituent and weighted on a per share basis. This was the only practically viable source for this purpose: Bloomberg does not provide the individual constituent weights required to build the estimate from the ground up, and dividend swap market data, which would represent the most market-consistent source of implied forward dividends, was not accessible. IN060 therefore represents the most granular and forward-looking dividend estimate available under these constraints.

The field has several properties that make it appropriate for the cost-of-carry framework. It is forward-looking rather than historical, making it conceptually suited to pricing a contract whose value depends on dividends yet to be paid. It is updated continuously as new information arrives, including earnings releases and corporate guidance, meaning it reflects the market’s evolving expectations at

each point in time. Crucially, historical values are available on a point-in-time basis, meaning each observation reflects only information available on that date, introducing no look-ahead bias into the back-test.

Construction of the Forward-Looking Dividend Yield

A futures contract priced at time d with expiry T is sensitive only to dividends expected to be paid between d and T (dividends already paid are sunk and have been incorporated into the current spot level). While we initially explored constructing the yield from the remaining dividend stream by subtracting year-to-date paid dividends from the estimated annual total and annualising over the fraction of the year remaining, this approach proved sensitive to timing mismatches near year-end and could occasionally produce implausible negative estimates. The dividend yield input to the fair value formula is therefore the Bloomberg-provided estimated dividend yield for the current fiscal year, converted directly to a continuously compounded basis:

$$q_d = \ln(1 + \text{Est. Dividend Yield}_d)$$

A cross-check confirms that the two approaches are consistent to within a tolerance of 0.01% across the full sample period outside of year-end boundary periods, validating the use of the simpler formulation. The resulting dividend yield series has a sample mean of approximately 1.71% per annum, a minimum of approximately 1.14%, and a maximum of approximately 2.79%, consistent with the observed historical range of the S&P 500 dividend yield.

Limitations of the Dividend Estimate

The primary limitation of this approach is transparency: Bloomberg does not disclose the full methodology underlying the IN060 estimate, making it difficult to assess precisely how constituent-level inputs are aggregated or how the estimate behaves in unusual market conditions. This opacity is an inherent constraint of using a proprietary data field rather than a fully reconstructible estimate.

Beyond this, consensus-based dividend estimates are subject to well-documented behavioral biases. Analyst projections tend to exhibit systematic optimism and herding, which can produce persistent directional errors, particularly around periods of corporate stress. Unusual corporate events such as dividend initiations, omissions, special distributions, and frequency changes, are also poorly captured, leading to larger forecast errors precisely when carry sensitivity is highest.

Finally, the relationship between corporate fundamentals and dividends is not stationary over time. Structural breaks around major market dislocations can cause historically calibrated estimates to systematically under perform, a risk that is particularly relevant for a back-test spanning the 2022 rate-hiking cycle and the associated shift from backwardation to contango.

Fair Value Computation

With all three inputs (the spot index level, the interpolated risk-free rate, and the forward dividend yield) estimated for each contract and each trading date, the fair value of each contract is computed as:

$$Fair_{c,d} = S_{0,d} \cdot \exp((r_{c,d} - q_d) \cdot T_{c,d})$$

This is applied across all 18 contracts over the full back-test period, generating 18 fair-value time-series, each active only during its respective contract's trading window. The computation is performed in vectorised form, applying the exponential function element-wise to the net-carry and time-to-expiry arrays.

Basis Analysis

The basis is defined as the difference between the observed market price of a futures contract and its computed fair value:

$$Basis_{c,d} = \text{Market Price}_{c,d} - \text{Fair Value}_{c,d}$$

Under the no-arbitrage condition, the basis should be zero. In practice, it fluctuates around zero as a result of temporary supply-demand imbalances, short-term funding pressures, dividend estimate revisions, and other market microstructure effects. A positive basis indicates the contract trades rich relative to fair value; a negative basis indicates it trades cheap.

The basis time-series serves two roles in the strategy. First, it provides a daily diagnostic of whether the current roll spread is fairly priced relative to theoretical carry: comparing the actual calendar spread between the front and next contract against the model-implied fair roll spread is the quantitative foundation of the roll decision. Second, and more directly, the basis is the input to the dynamic rolling signal described in Section 5.

Roll Signal: The Basis Z-Score

The dynamic rolling strategy is triggered by a statistical signal derived from the basis. Rather than acting on the level of the basis directly, which would be sensitive to the prevailing interest rate and dividend regime and therefore not comparable across different periods, the signal is normalized into a rolling z-score. For each trading date d , the z-score of the basis is computed over a look-back window of N trading days as:

$$z_d = \frac{Basis_d - \mu_{d,N}}{\sigma_{d,N}}$$

where $\mu_{d,N}$ is the rolling mean and $\sigma_{d,N}$ is the rolling standard deviation of the basis over the N days preceding date d . The z-score expresses the current basis in units of its own recent historical volatility, rendering the signal dimensionless and directly comparable across market regimes with different absolute spread levels. A z-score of for example +2, indicates that the current basis is two standard deviations above its recent average, the contract is anomalously expensive relative to fair value, suggesting that the roll spread has widened beyond what carry fundamentals justify.

The strategy uses this signal as follows: when the z-score exceeds a positive threshold within a pre-defined window before expiry, an early roll is executed, capturing the rich spread before it mean-reverts. If no signal is triggered within the window, a forced roll is executed mechanically a fixed number of days before expiry, ensuring that the position is never held to settlement. The precise parameter choices, look back window N and z-score threshold, are determined through a grid search described in Section 5.

Data Pipeline Summary

The complete data pipeline underlying this section proceeds through five stages:

1. **Futures prices:** daily closing prices for all 18 quarterly E-Mini contracts over their respective active periods, sourced from Bloomberg.
2. **Spot index:** daily S&P 500 closing prices from Yahoo Finance, used as S_0 in the fair value formula.
3. **Risk-free rates:** daily CME Term SOFR at four standard tenors (1M, 3M, 6M, 12M), extracted from Bloomberg, merged into a single time-series panel, and converted from percentage to decimal.
4. **Dividend yield:** Bloomberg IN060 field (IDX_EST_DVD_CURR_YR), providing the index-level estimated annual dividends for the S&P 500 for the current fiscal year, converted to a continuously compounded rate.
5. **Fair value and basis:** computed analytically from the four inputs above, using the cost-of-carry formula with cubic-spline-interpolated rates matched to each contract's time-to-expiry.

Roll event log: a daily record of the near and far contract pair, the market roll spread, the fair roll spread, their difference, and the basis z-score, produced as a structured output and used as the primary input to the optimised strategy analysis presented in Section 5.

4 BASELINE STRATEGY: SYSTEMATIC ROLLING

The objective of this segment is to create a basic rolling strategy for E-mini S&P 500 futures to evaluate more sophisticated strategies by comparing their results with those of this base strategy. The rolling strategy used is purely mechanical and ignores market information, allowing the impact of the rolling mechanism, transaction costs, and interest rates to be evaluated separately.

Systematic Rolling Protocol

As futures contracts have a finite maturity, investors will need to roll over their positions regularly to maintain market exposure. For our baseline rolling strategy, rolling is conducted systematically based on time-to-maturity. At each date, the strategy holds the front contract and rolls into the next one when the remaining time to maturity falls below a predefined threshold. In other words, to ensure continuous exposure, this involves closing the contract near its expiration and opening an equivalent position in a contract with a longer maturity.

The strategy maintains a constant short position in E-mini S&P 500 futures, consistent with a hedging framework. The exposure remains unchanged over time, as rolling only affects the contract's maturity, not its size or direction.

The back-test incorporates realistic trading frictions, including a \$50 contract multiplier (obtained from the CME Group website) and an assumed transaction cost of \$2.50 per trade. This assumption is consistent with all-in institutional rates for E-Mini S&P 500 futures contracts, which typically range from \$1.50 to \$3.00 per contract for high-volume market participants, including exchange fees, clearing fees, and brokerage commissions. Since each roll involves two trades, these costs accumulate over time and impact performance. This setup corresponds to a systematic short 1-contract strategy, where each roll consists of buying back the near contract and shorting the next one. As a result, the profitability of the roll depends on the shape of the term structure: in a contango environment ($r > q$), the strategy tends to generate positive roll yield, whereas in backwardation ($r < q$), it leads to losses.

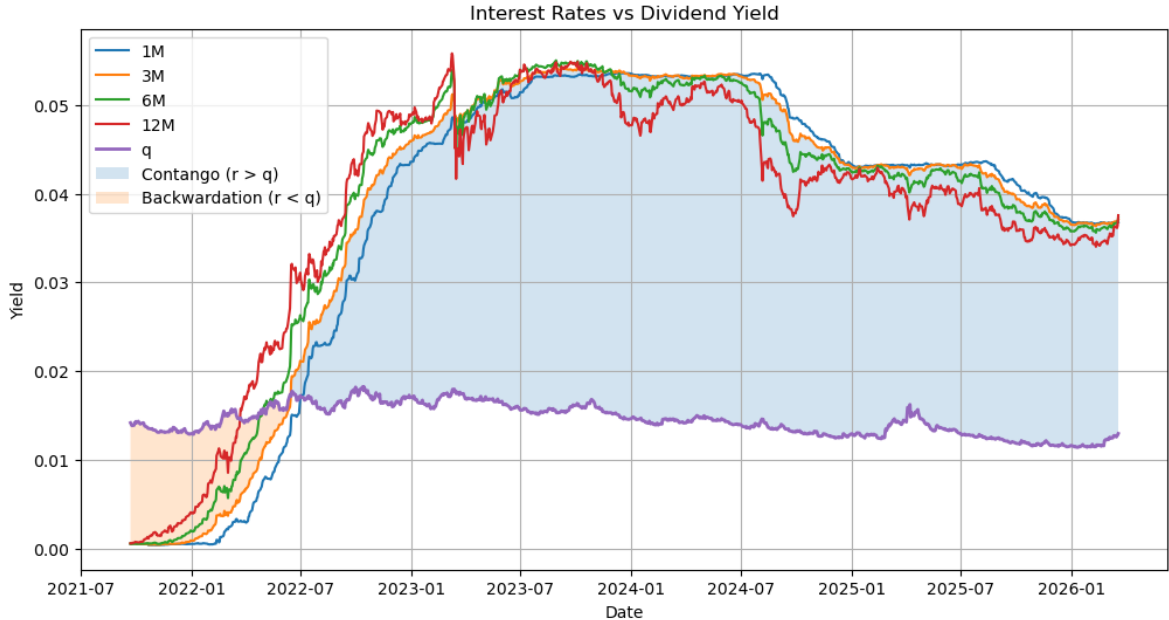


Figure 1: Interest Rate " r " versus Dividend Yield " q "

Daily PnL is computed from price variations, the short position, and the contract multiplier. Transaction costs are deducted to obtain net PnL, which is then aggregated to compute the portfolio's NAV.

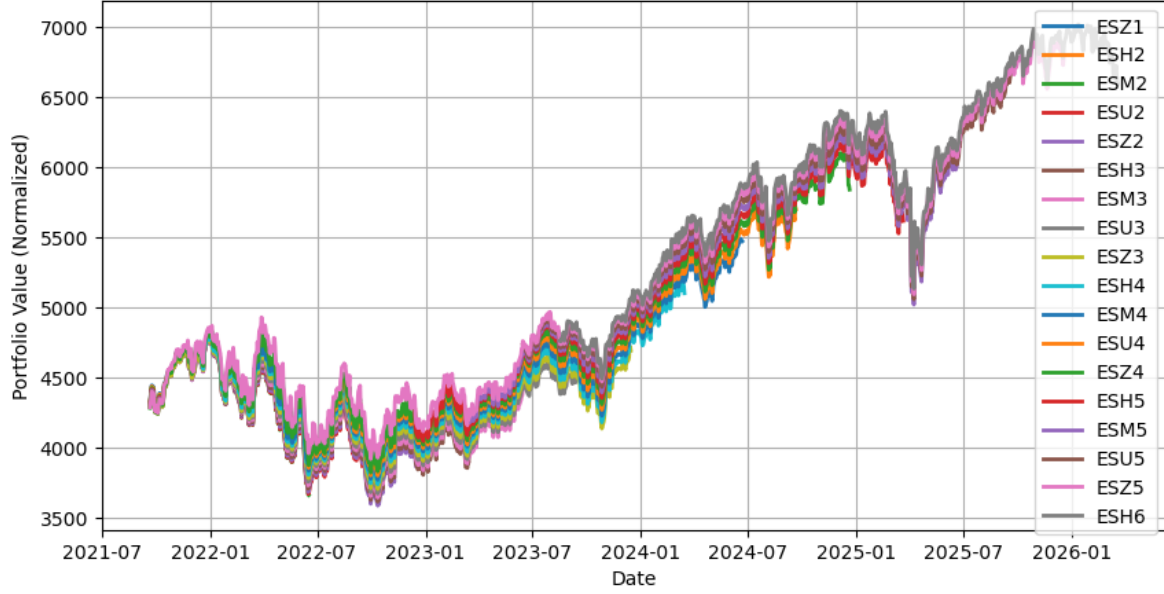


Figure 2: Futures Prices comparison

A key component of the analysis is the comparison between market roll spreads and fair value roll spreads. Their difference captures mispricing in the term structure and represents the economic impact of rolling.

Simulation Parameters & Sensitivity Analysis

To measure the impact of roll timing, we test using rolling periods ranging from $T-1$ to $T-10$ business days before expiration. This helps us identify the most sensitive period in this strategy's performance relative to other periods.

Rolling Day	Final NAV	Market Rolling Spread	Fair Rolling Spread	Difference
$T-10$	99,921,833.5	689.75	558.62	131.12
$T-9$	99,921,646.0	686.00	553.08	132.91
$T-8$	99,921,571.0	684.50	552.24	132.25
$T-7$	99,921,658.5	686.25	556.14	130.10
$T-6$	99,921,671.0	686.50	557.97	128.52
$T-5$	99,922,196.0	697.00	556.13	140.86
$T-4$	99,922,533.5	703.75	557.60	146.14
$T-3$	99,922,421.0	701.50	555.45	146.04
$T-2$	99,921,758.5	688.25	557.92	130.32
$T-1$	99,919,298.0	639.04	557.79	81.24

Table 2: Static Rolling Analysis Data

Performance Metrics & Key Insight

Over the full backtest period (21 September 2021 to 20 March 2026), the baseline strategy executed 17 rolls in total, one per quarterly contract expiry. The results show that the baseline strategy consistently delivers slightly negative performance, with a total return of approximately **-0.08%** across all rolling configurations. The final NAV remains close to the initial value of \$100 million, ranging between **\$99.919m** and **\$99.922m**. This indicates that the strategy is roughly break-even but slightly

loss-making after costs.

The Sharpe ratio is also negative, at about **-0.42**. That reflects poor risk-adjusted performance, suggesting that a purely mechanical roll strategy yields no gains.

Nonetheless, when the time for rolling is taken into account, the effect on performance is only slight. While small variations exist (for example, a final NAV of **\$99.919m** for *T-1* versus **\$99.922m** for *T-4*), these differences remain marginal. This indicates that improving a fixed rolling schedule alone is insufficient to materially improve performance.

However, looking at roll spreads, some important conclusions can be made. The cumulative market roll spread is always greater than the fair-value roll spread. For instance, in the *T-10* configuration, the total market roll amounts to approximately 689.75, compared to a fair roll of 558.62, resulting in a difference of about 131.13.

Across all configurations, this difference ranges from roughly +80 to +145. This implies disparities between the market price and the theoretical value. From the findings, it can be concluded that there are systematic inefficiencies in the futures market that cannot be explained by a naive rolling approach.

Strategic Constraints & Model Weaknesses

Despite providing a useful benchmark, this strategy has important limitations. The rolling decision is entirely static and time-based, ignoring any information about market conditions or pricing variations from fair value. As a result, the strategy cannot respond to changing market conditions or exploit temporary mispricing.

Furthermore, even though different rolling windows are tested, the improvements remain marginal, highlighting the limits of optimizing a fixed schedule. The strategy fails to exploit the observed differences between market and fair value roll spreads.

These limitations motivate a more sophisticated approach, in which rolling decisions are driven by dynamic signals derived from market inefficiencies rather than predetermined rules.

5 OPTIMIZED STRATEGY: DYNAMIC ROLLING

The conventional approach is to roll mechanically. The proposed alternative is to roll opportunistically: By taking advantage of mispricing in the next contract relative to its theoretical fair value. The objective is to demonstrate that this smarter timing produces a measurable, repeatable improvement over the baseline strategy.

The Approach

The approach has three moving parts. The first two define what counts as a valid trading signal, the third turns that signal into action.

The signal: where mispricing comes from

Each day, two numbers are compared: the market spread and the fair spread. The difference between them, the basis, is the raw signal. A persistently positive basis indicates the market is overpaying for the next contract; a temporary spike indicates a short window of opportunity to roll.

The Z-score: making “unusual” measurable

Raw basis values drift over time as interest rates, dividends, and term-structure regimes change. A fixed threshold (“roll whenever basis exceeds X”) would be unstable. Instead, the basis is standardized to a Z-score, a measure of how unusual today’s value is relative to its recent history. A Z-score above 3 means today’s mispricing is more than three standard deviations beyond the recent norm: a genuine outlier.

The decision rule

Each day, two simple checks run in order:

- **Signal Roll:** if expiry is within 14 calendar days and the Z-score has crossed the threshold, roll early to capture the mispricing.
- **Forced Roll:** if no signal has fired and expiry is within 3 days, roll anyway. The hedge is never allowed to expire unrolled.

This two-tier design keeps the hedge intact at all times while leaving room for opportunistic timing. The strategy uses the Z-score of yesterday when making the decision, so no decision is ever based on data that would not be available in real time.

Parameter Selection

Two knobs control the strategy: how many days of history to use when computing the Z-score (the look-back), and how unusual the score must be to trigger a roll (the threshold). All nine combinations of the values $\{10, 20, 30 \text{ days}\} \times \{2.0, 2.5, 3.0\}$ were tested.

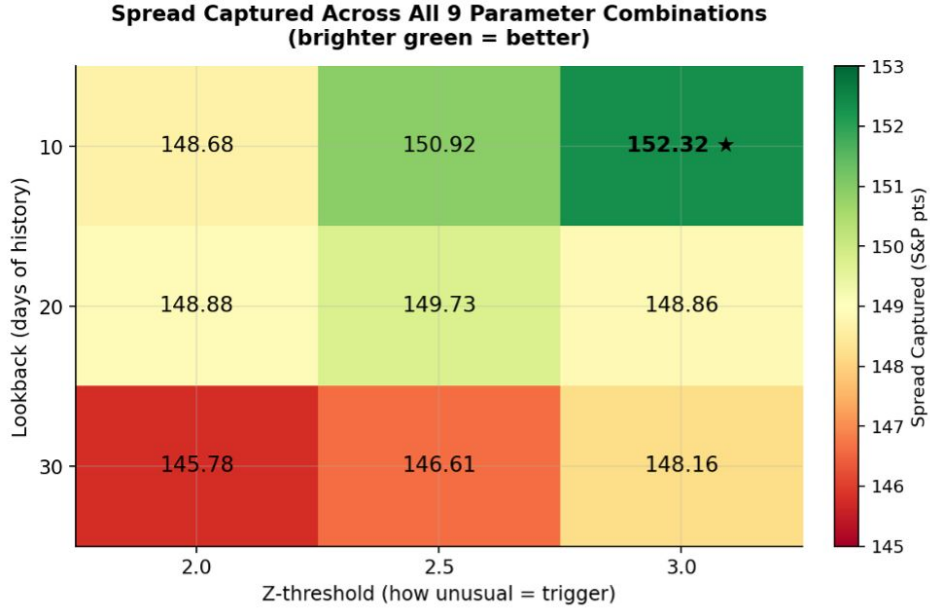


Figure 3: Spread captured for each combination of look-back window and Z-threshold

The optimal region is in the top-right corner: short look-back combined with a high threshold.

- A short look-back means the Z-score adapts quickly when market regimes change. Older data becomes misleading once conditions shift.
- A high threshold filters out small mispricing that are likely just noise. Acting only on truly extreme moves preserves the quality of every signal-driven roll.

Spread capture varies by less than 5% across all nine configurations. Indicating that the result holds across a broad combination of parameter choices, not only at one sweet spot.

Key Results

Across nine parameter combinations of the optimized strategy, eight performed better than the best fixed-window baseline. The single best configuration improved spread capture by 4.22%, equivalent to more than 6 additional S&P 500 index points per contract during the back-test period.

Metric	Best Fixed Baseline	Best Optimized	Improvement
Spread Captured (S&P pts)	146.15	152.32	+4.22%
In USD per contract	\$ 7,307.50	\$ 7,616.00	\$ 308.50
Final NAV (on \$100M)	\$ 99,922,533	\$ 99,922,896	\$ 363
Configs beating baseline	—	8 of 9	—
Total roll events	17 (all forced)	17 (10 signal + 7 forced)	—

Table 3: Performance Comparison: Fixed Baseline vs. Optimized Configurations

Strategy Comparison

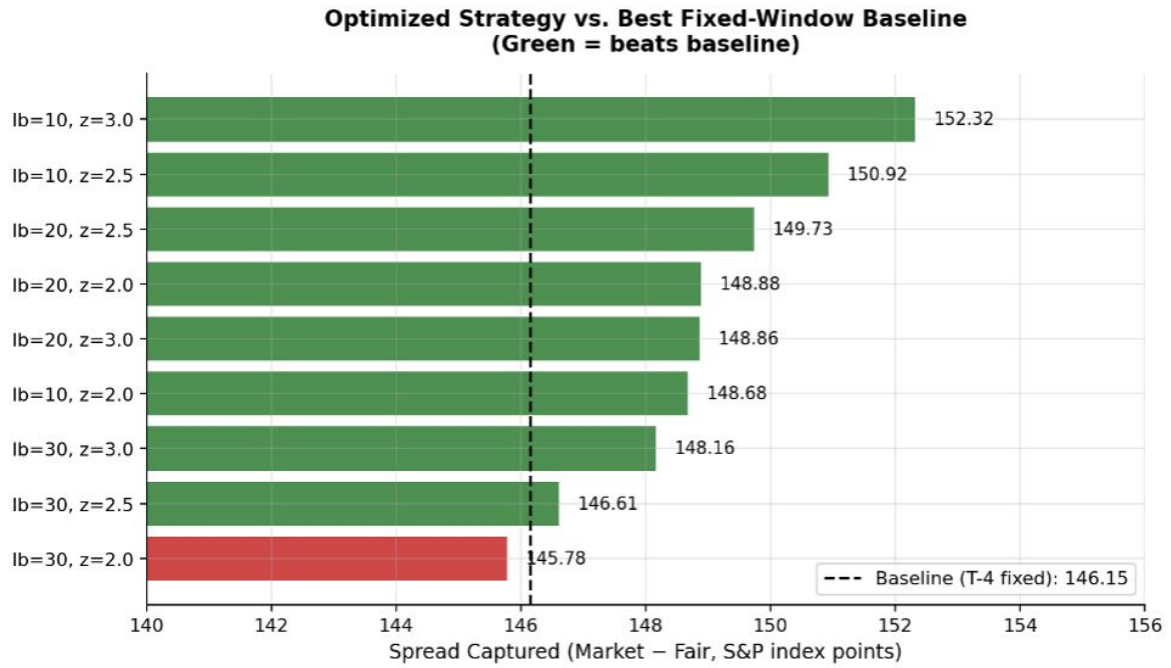


Figure 4: Spread captured by every parameter combination of the optimized strategy, compared against the best fixed-window baseline

The vertical dashed line is the best result the fixed-schedule strategy could produce (rolling at T-4 days). Every green bar represents a configuration of the optimized strategy that beat that benchmark. Only one configuration (a long look-back combined with a low threshold) underperformed it slightly.

6 BACK-TEST RESULTS AND PERFORMANCE EVALUATION

Since the total market roll yield and the market-fair yield spread are robust across different look-back windows and z -score thresholds, we choose a 20-day look-back window and a z -score threshold of 2.0 to back-test our systematic futures shorting strategy combined with our equal-weight long portfolio.

To determine the number of futures contracts required for hedging, we monitor the daily NAV of our long portfolio and estimate the portfolio's rolling beta relative to the S&P 500 using a 60-day rolling window. The hedge ratio h controls the degree of hedging; $h = 0$ represents no hedge, and $h = 1$ represents a fully hedged portfolio. The number of futures contracts is calculated as:

$$N_t = h \cdot \frac{\beta_{t-1} \cdot I_{t-1}}{F_t \cdot M}$$

where β_{t-1} is the lagged rolling beta estimate, I_{t-1} is the previous day's portfolio NAV, F_t is the futures price at time t , and M is the futures contract multiplier. We use lagged beta and portfolio NAV to avoid look-ahead bias, while the current futures price is used because the hedge is entered at time t .

We start with an initial NAV of USD 100 million and assume an initial transaction cost of 20 bps for purchasing our equal-weight portfolio, plus transaction costs of \$2.50 per futures contract per side. We do not rebalance the hedge daily because daily hedge adjustments would generate substantial transaction costs. Instead, the hedge size is recalculated only when the futures position is rolled into the next contract. We also test performance across no-hedge (the equal-weight portfolio), quarter-hedge, half-hedge, and full-hedge strategies to compare our portfolio's performance.

Strategy comparison

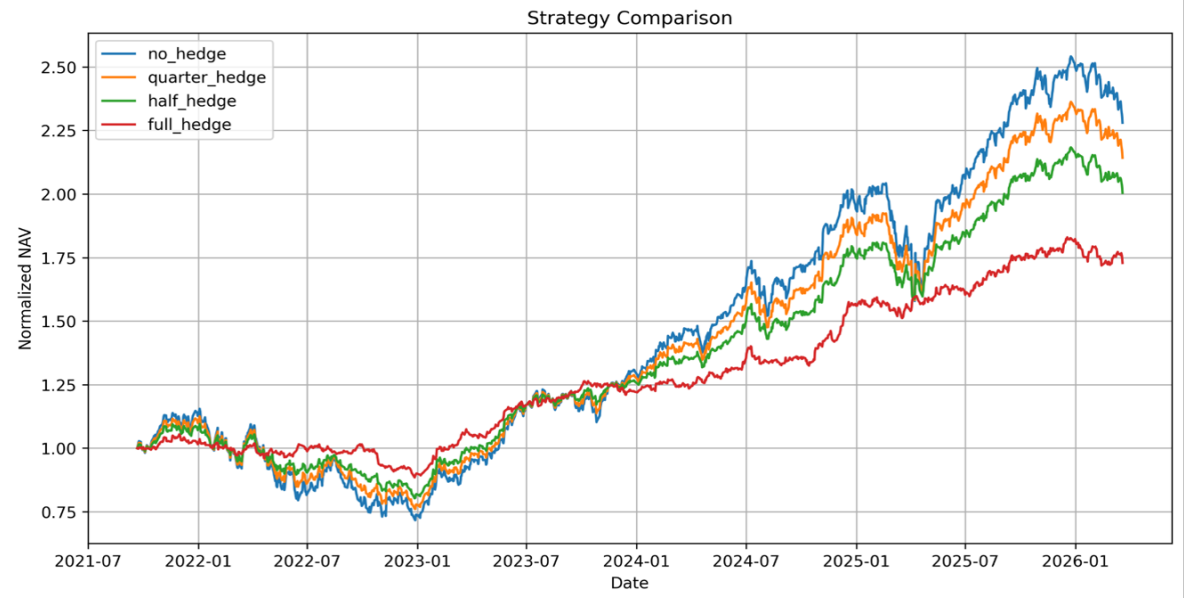


Figure 5: Cumulative returns for different level of hedged portfolio

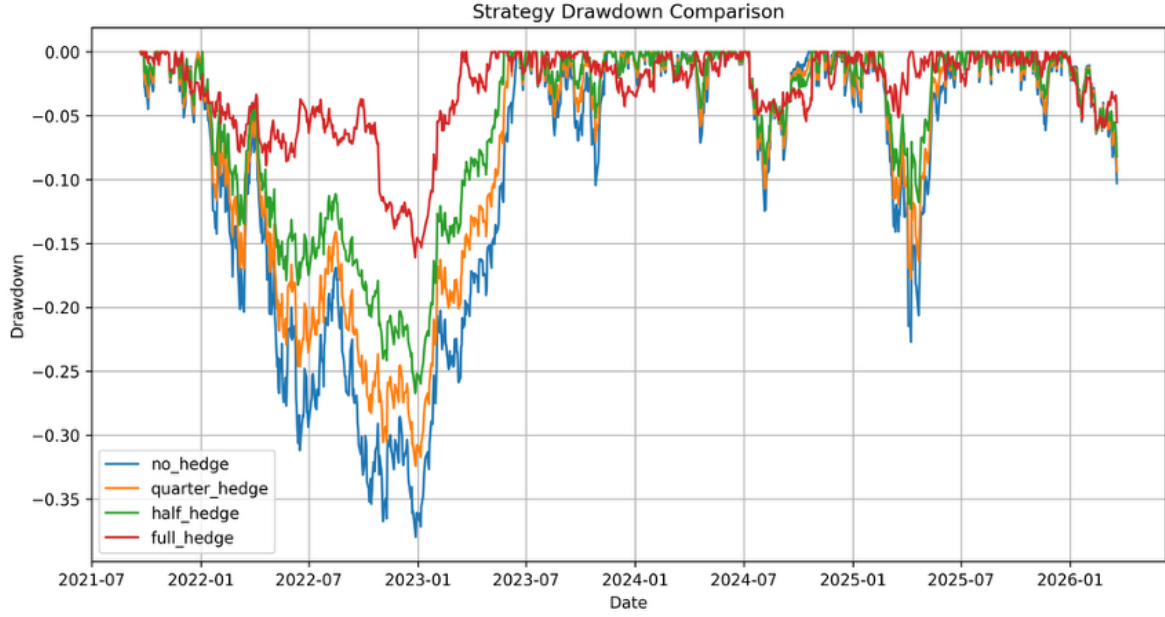


Figure 6: Strategy Draw Down comparison

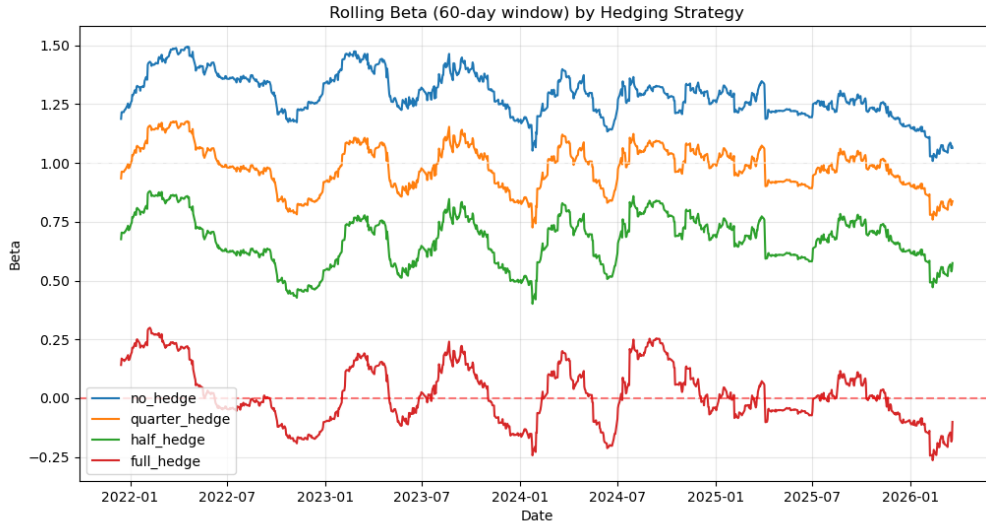


Figure 7: Rolling Beta for different level of Hedge

Table 4: Hedging Strategy Performance Metrics

Strategy	Annualized Return	Annualized Volatility	Sharpe Ratio	Calmar Ratio	Max Drawdown	Win Rate
<i>no_hedge</i>	0.202	0.242	0.880	0.532	-0.380	0.539
<i>quarter_hedge</i>	0.185	0.191	0.985	0.571	-0.324	0.532
<i>half_hedge</i>	0.168	0.145	1.141	0.628	-0.267	0.523
<i>full_hedge</i>	0.130	0.096	1.318	0.807	-0.161	0.539

The equal-weight portfolio without hedging achieved the highest return and final NAV, while increasing the hedge ratio reduced portfolio returns. Final NAV declined from 227.5M in the unhedged

Table 5: Risk Distribution and Cost Analysis

Strategy	Max Daily Return	Min Daily Return	Skewness	Kurtosis	Final NAV	Total Trans. Cost
<i>no_hedge</i>	0.120	-0.061	0.254	4.757	227,518,671.752	200,000.000
<i>quarter_hedge</i>	0.090	-0.046	0.207	3.723	213,768,654.474	215,033.304
<i>half_hedge</i>	0.060	-0.039	0.159	2.520	200,018,637.195	230,066.608
<i>full_hedge</i>	0.033	-0.025	0.040	1.740	172,518,602.639	260,133.216

portfolio to 213.8M, 200.0M, and 172.5M as the hedge ratio increased. Annualized return also decreased from 20.2% to 18.5%, 16.8%, and 13.0%. Hedging plays a key role in risk management, particularly during drawdown periods. For example, during the 2022 stock market decline, the no-hedge portfolio suffered the most severe losses due to its exposure to U.S. equities, while the fully hedged portfolio suffered the least.

Although the fully hedged strategy has the lowest return and final NAV, it has the highest Sharpe ratio (1.318) and Calmar ratio (0.807) among the other three portfolios. This is caused by its lowest volatility (9.6% annualized) and much smaller drawdowns (max drawdown = -16.1%).

The initial transaction cost of the long portfolio is 200,000, while the remaining transaction costs are the total transaction costs generated from each futures rolling. As expected, the more we hedge, the more we pay for the transaction costs.

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- [3] Brown, L., Call, A., Clement, M., and Sharp, N. (2002). “Inside the “black box” of sell-side financial analysts.” *Journal of Accounting Research*, 53(1), pp. 1–47.